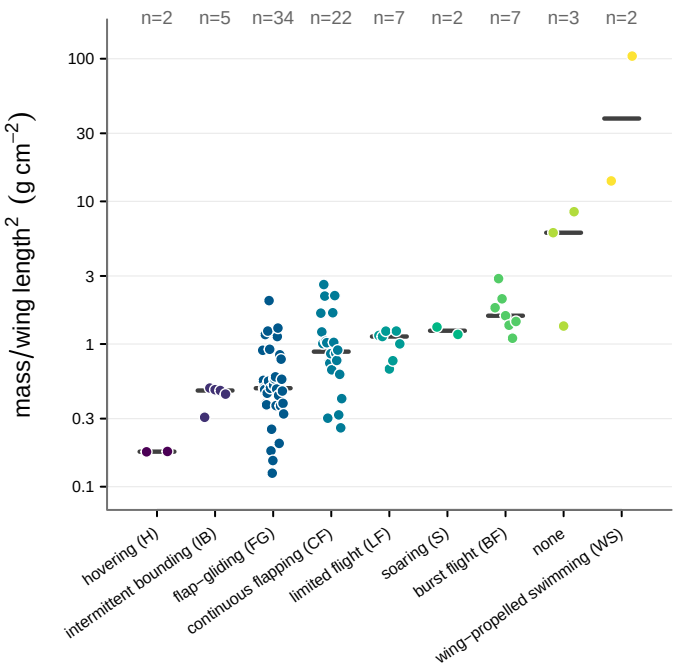
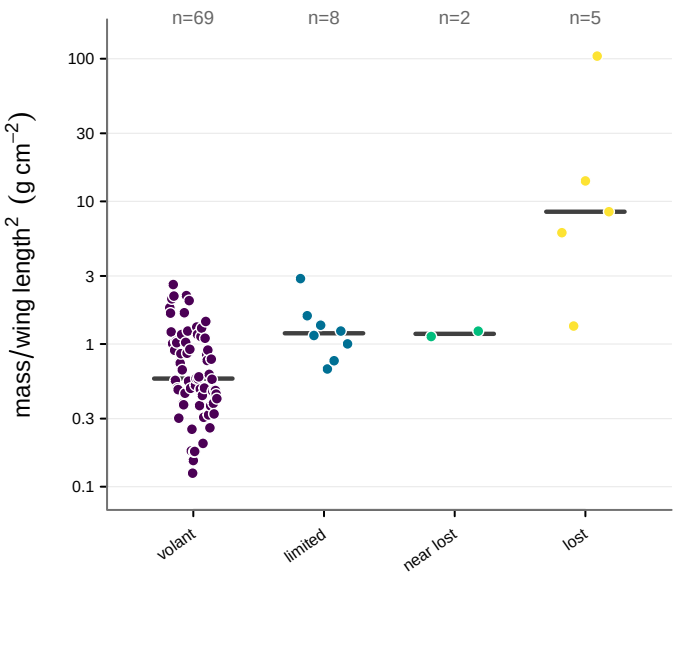
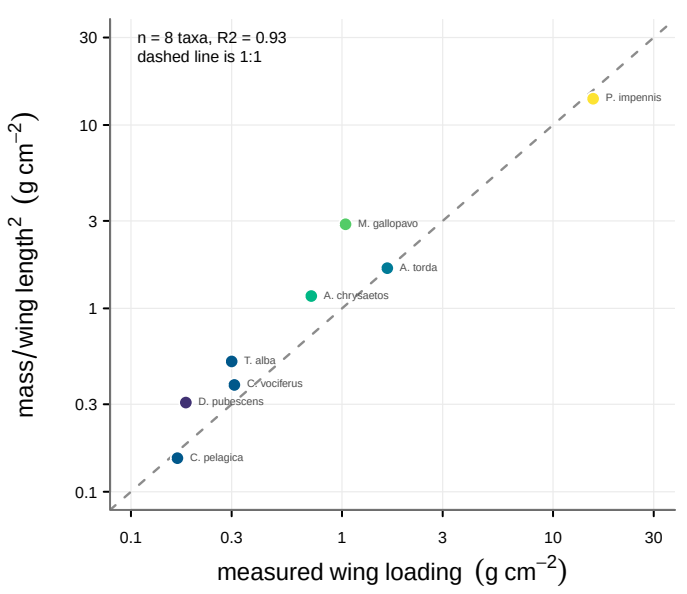
A


n=2
n=5
n=34
n=22
n=7
n=2
n=7
n=3
n=2
mass/wing length^2 (g cm^{-2})
100
30
10
3
1
0.3
0.1
hovering (H)
intermittent bounding (IB)
flap−gliding (FG)
continuous flapping (CF)
limited flight (LF)
soaring (S)
burst flight (BF)
none
wing−propelled swimming (WS)

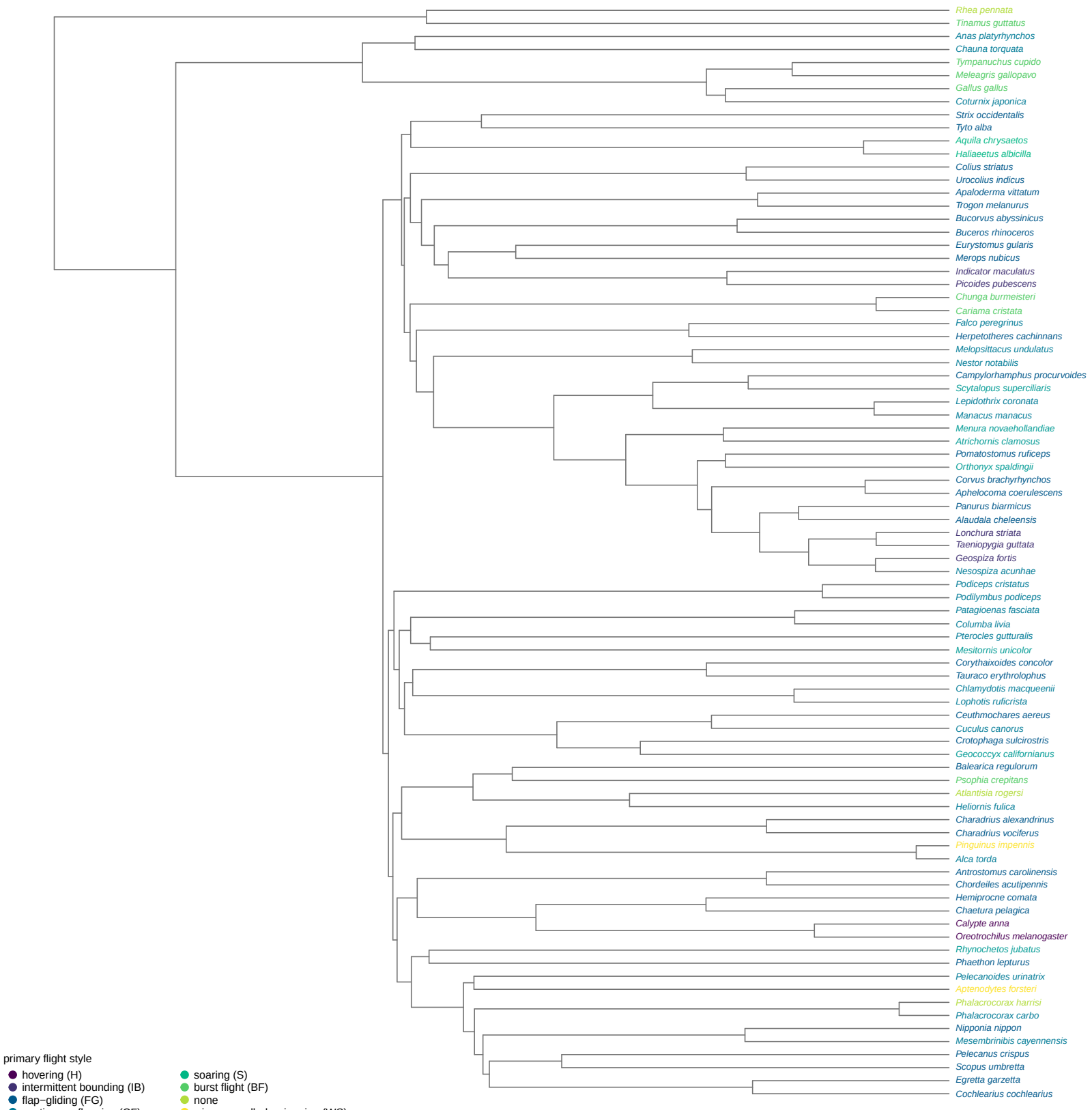
B


n=69
n=8
n=2
n=5
mass/wing length^2 (g cm^{-2})
100
30
10
3
1
0.3
0.1
volant
limited
near lost
lost

C


n = 8 taxa, R2 = 0.93
dashed line is 1:1
mass/wing length^2 (g cm^{-2})
30
10
3
1
0.3
0.1
P. impennis
M. gallopavo
A. torda
A. chrysaetos
T. alba
C. vociferus
D. pubescens
C. pelagica
measured wing loading (g cm^{-2})

D


Rhea pennata
Tinamus guttatus
Anas platyrhynchos
Chauna torquata
Tympanuchus cupido
Meleagris gallopavo
Gallus gallus
Coturnix japonica
Strix occidentalis
Tyto alba
Aquila chrysaetos
Haliaeetus albicilla
Colius striatus
Urocolius indicus
Apaloderma vittatum
Trogon melanurus
Bucorvus abyssinicus
Buceros rhinoceros
Eurystomus gularis
Merops nubicus
Indicator maculatus
Picoides pubescens
Chunga burmeisteri
Cariama cristata
Falco peregrinus
Herpetotheres cachinnans
Melopsittacus undulatus
Nestor notabilis
Campylorhamphus procurvoides
Scytalopus superciliaris
Lepidothrix coronata
Manacus manacus
Menura novaehollandiae
Atrichornis clamosus
Pomatostomus ruficeps
Orthonyx spaldingii
Corvus brachyrhynchos
Aphelocoma coerulescens
Panurus biarmicus
Alaudala cheleensis
Lonchura striata
Taeniopygia guttata
Geospiza fortis
Nesospiza acunhae
Podiceps cristatus
Podilymbus podiceps
Patagioenas fasciata
Columba livia
Pterocles gutturalis
Mesitornis unicolor
Corythaixoides concolor
Tauraco erythrolophus
Chlamydotis macqueenii
Lophotis ruficrista
Ceuthmochares aereus
Cuculus canorus
Crotophaga sulcirostris
Geococcyx californianus
Balearica regulorum
Psophia crepitans
Atlantisia rogersi
Heliornis fulica
Charadrius alexandrinus
Charadrius vociferus
Pinguinus impennis
Alca torda
Antrostomus carolinensis
Chordeiles acutipennis
Hemiprocne comata
Chaetura pelagica
Calypte anna
Oreotrochilus melanogaster
Rhynochetos jubatus
Phaethon lepturus
Pelecanoides urinatrix
Aptenodytes forsteri
Phalacrocorax harrisi
Phalacrocorax carbo
Nipponia nippon
Mesembrinibis cayennensis
Pelecanus crispus
Scopus umbretta
Egretta garzetta
Cochlearius cochlearius
primary flight style
hovering (H)
intermittent bounding (IB)
flap−gliding (FG)
continuous flapping (CF)
limited flight (LF)
soaring (S)
burst flight (BF)
none
wing−propelled swimming (WS)